



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Course: CSE 391

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Project Title: AutoTransport — an online vehicle shipping platform with a Laravel REST API, a staff admin panel and a React Native mobile application

Project Features:

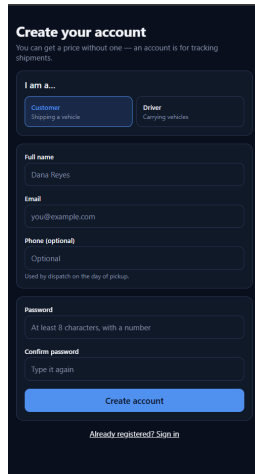
The following features were designed and implemented for this project:

1. Authentication and accounts — registration for both customer and driver account types, login, logout and password reset. The mobile client authenticates with Laravel Sanctum bearer tokens held in the device Keychain (iOS) / Keystore (Android).
2. Role-based access control — six roles (super-admin, admin, dispatcher, support, driver, customer) and fourteen permissions, editable from the admin panel without a redeploy. The mobile app routes each role to its own panel.
3. Service catalogue and instant estimate — seven transport services with base price, per-mile rate, minimum price and transit estimates; an unauthenticated estimator prices a lane before the customer signs up.
4. Booking — a customer books a shipment from the app, choosing the transport option, addresses, collection date and up to eight vehicles. The request writes the full quote request, quote and booking chain in a single database transaction so the pricing history is preserved.
5. Shipment tracking — an append-only event timeline. Internal dispatch notes share the same table and are filtered out of the customer-facing API.
6. Driver panel — a driver sees only the loads assigned to them and reports progress (picked up, in transit, delivered), with an optional GPS position recorded as an internal event.
7. Admin panel — server-rendered dashboard with an awaiting-confirmation queue, dispatch board, carrier / driver / truck assignment, manual payment recording, review moderation, user and role management, service catalogue editing, settings and a contact inbox.
8. Review and rating after service — a customer rates a delivered shipment overall plus four optional sub-ratings (communication, timeliness, vehicle condition, value). Nothing is published until a moderator approves it, and service rating averages are maintained automatically.
9. Payments — an append-only payment ledger with duplicate-submission protection; recording the full balance confirms the booking automatically.

10. Contact form — a public, rate-limited enquiry form feeding the staff inbox.
11. Automated test suite — 133 feature tests and 623 assertions covering authentication, booking, the review flow, the driver API, payments and admin access control.

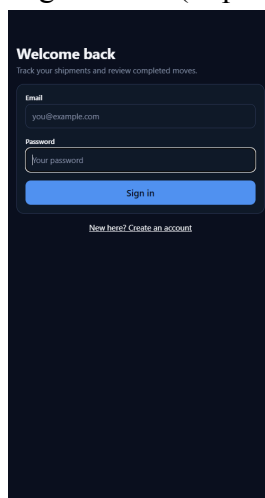
Screenshots:

12. Registration screen showing the Customer / Driver account type selector, with the driver licence fields revealed (<http://localhost:8081/register>).



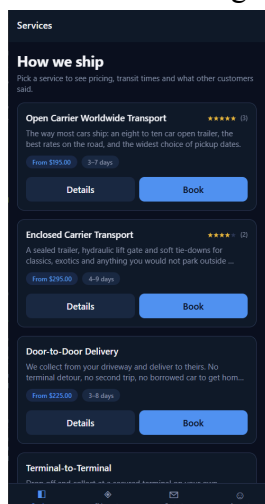
The screenshot shows a registration screen titled "Create your account". It features a selector for "I am a..." with two options: "Customer" (Shipping vehicles) and "Driver" (Carrying vehicles). Below this are input fields for "Full name" (Dana Reyes), "Email" (you@example.com), and "Phone (optional)" (Optional). A note indicates the phone is "Used by dispatch on the day of pickup". There are also fields for "Password" (At least 8 characters, with a number) and "Confirm password" (Type it again). A blue "Create account" button is at the bottom, with a link "Already registered? Sign in" below it.

13. Login screen (<http://localhost:8081/login>).



The screenshot shows a login screen titled "Welcome back". It features input fields for "Email" (you@example.com) and "Password" (Your password). A blue "Sign in" button is below the password field. At the bottom, there is a link "New here? Create an account".

14. Services tab listing the transport options with Details and Book actions.



The screenshot shows the "Services" tab with the title "How we ship". It lists three transport options, each with a "Details" and "Book" button:

- Open Carrier Worldwide Transport** (★★★★ (1))
The way most cars ship: an eight to ten car open trailer, the best rates on the road, and the widest choice of pickup dates.
From \$395.00 | 3-7 days
- Enclosed Carrier Transport** (★★★★ (2))
A sealed trailer, hydraulic lift gate and soft tie-downs for classics, exotics and anything you would not park outside ...
From \$295.00 | 4-9 days
- Door-to-Door Delivery**
We collect from your driveway and deliver to theirs. No terminal fee, no second trip, no borrowed car to get home.
From \$225.00 | 3-8 days

At the bottom, there is a section for "Terminal-to-Terminal" with a note: "Please call our toll-free line to understand terminal-to-terminal cost and service."

15. Booking form showing the transport option chips, addresses and the vehicle "starts and drives" toggle.

The first screen, 'Book a shipment', displays transport options: 'Open Carrier Worldwide Transport' (selected), 'Enclosed Carrier Transport', 'Door-to-Door Delivery', 'Terminal-to-Terminal', 'Expedited Transport', 'Motorcycle Transport', and 'Heavy Equipment Transport'. It also shows collection and delivery addresses in Austin, TX. The second screen, 'Book a shipment', shows vehicle selection with options like Sedan, Coupe, SUV, Pickup Truck, Minivan, Motorcycle, and Heavy/Oversize. It includes a 'starts and drives' toggle and a 'Confirm booking' button. The third screen, 'Shipment', shows the confirmed booking details for 'Dhaka → Chattogram', including the schedule, payment status, and tracking history.

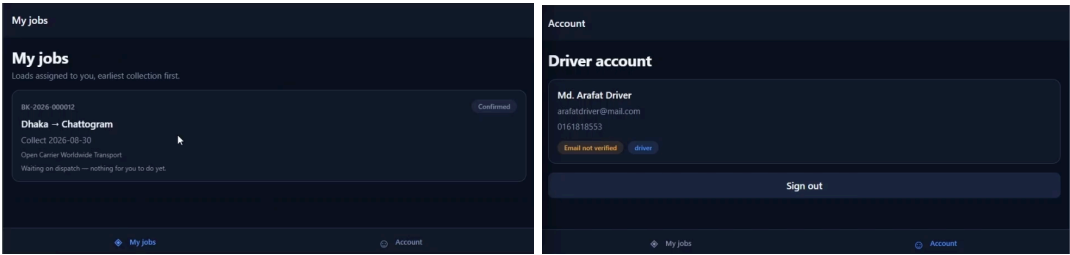
16. Shipments tab with a delivered booking offering the Leave a review action.

The first screen, 'Shipment', shows the details for a delivered shipment 'Dhaka → Chattogram'. It includes the schedule, payment status, and tracking history. A 'Leave a review' button is visible. The second screen, 'Leave a review', shows the review form with a star rating and a text input field for the review.

17. Review screen showing the overall rating and the four sub-ratings.

The screen shows the review details for a shipment. It includes the overall rating (5.0 stars) and four sub-ratings: Communication (5 stars), Timeliness (5 stars), Vehicle condition (4 stars), and Value for money (4 stars). The review text is 'There is a Mark but it wiped off'. The review is from 'Mid. A. Open Carrier Worldwide Transport' and was recorded on '2025-08-22 12:41'. The review is currently 'Pending'.

18. Driver panel — My jobs, showing an assigned load and its next-step action.



19. Admin dashboard showing the Awaiting confirmation queue (http://127.0.0.1:8000/admin).

The image shows a screenshot of an Admin dashboard for 'Bookings'. The dashboard includes a search bar, filters for status and pickup location, and a table of bookings. The table has columns for Booking, Customer, Lane, Pickup, Status, and Total. The bookings are listed in descending order of total amount.

BOOKING	CUSTOMER	LANE	PICKUP	STATUS	TOTAL
BK-2026-000012	Md. Arafat arafatsarkar@gmail.com	Dhaka, DH → Chattogram, CU 1,000 mi	Sun 30 Aug 2026	Awaiting Payment	USD 815.00
BK-2026-000011	Arafat Sarkar arafat@gmail.com	Dhaka, DH → Chattogram, CU 1,000 mi	Mon 31 Aug 2026	Awaiting Payment	USD 815.00
BK-2026-000010	Dana Whitfield customer@autotransport.test	Austin, TX → Denver, CO 1,000 mi	Thu 27 Aug 2026	Confirmed	USD 815.00
BK-2026-000009	Deleted user	Dhaka, NO → Dhaka, NO 1,000 mi	Sun 30 Aug 2026	Confirmed	USD 1,467.00
BK-2026-000008	Dana Whitfield customer@autotransport.test	Austin, TX → Denver, CO 1,000 mi	Sat 22 Aug 2026	Confirmed	USD 2,750.00

20. Admin booking detail showing the status form, the assignment form and the Record a payment form.

Admin Dashboard

Operational snapshot for Saturday 22 August 2026.

OPEN LEADS
0
New, reviewing or quoted

AWAITING CONFIRMATION
1
New orders, not yet confirmed

ACTIVE SHIPMENTS
3
Confirmed through in transit

AWAITING MODERATION
1
Nothing is public until approved

UNREAD MESSAGES
0
Contact form, never opened

CAPTURED (30 DAYS)
\$11,572.40
Captures less refunds and chargebacks

ACTIVE ACCOUNTS
12
Customers, drivers and staff

AWAITING CONFIRMATION 1

View all

ORDER	CUSTOMER	LANE	PLACED
BK-2026-000011	Arafat Sarkar	Dhaka -- Chattogram	1 day ago

Admin Bookings

BK-2026-000012

TIMELINE

Internal entries included

Booked Pending Payment

Booked in the app against Open Carrier Worldwide Transport at the automated estimate (USD 815.00). Awaiting payment.

21 Aug 2026 13:57 by Md. Arafat

Status Change Confirmed

Status changed to Confirmed.

21 Aug 2026 13:57 by Site Administrator

Payment Recorded

Full of USD 815.00 recorded by Site Administrator — Got it.

21 Aug 2026 13:59 by Site Administrator

Assignment Changed Internal

Assigned carrier: Lone Star Auto Logistics, driver: Md. Arafat Driver, truck: 118.

21 Aug 2026 13:59 by Site Administrator

Status Change Assigned

Status changed to Carrier Assigned.

22 Aug 2026 12:41 by Site Administrator

Status Change Picked Up

Status changed to Picked Up.

22 Aug 2026 12:43 by Site Administrator

Status Change In Transit

Status changed to In Transit.

22 Aug 2026 12:43 by Site Administrator

Status Change Delivered

Status changed to Delivered.

22 Aug 2026 12:43 by Site Administrator

Admin Bookings

BK-2026-000012

MOVE STATUS

Delivered is terminal — there is nowhere left to move.

MONEY

TOTAL
USD 815.00

DEPOSIT
USD 163.00

PAID
USD 815.00

BALANCE
USD 0.00

Ledger

Captured Full 815.00

Refunds are new rows, never edits to a capture (\$4.11).

RECORD A PAYMENT

Amount (USD) *

0.00

Pre-filled with the outstanding balance. Entered in USD, stored as integer cents.

Type *

Balance

Method *

Bank transfer

Reference

21. Admin review moderation queue with a pending review.

Admin Reviews

Reviews

Nothing reaches the public site or a rating average until it is approved here.

All 6 Pending 1 Approved 5 Rejected 0

★★★★★ 5.0 Good Pending

There is a Mark but it wiped off

Md. A. Open Carrier Worldwide Transport BK-2026-000012 11 minutes ago

Approve Reject Feature

Open

1 result

Admin Reviews Good

Review by Md. A.

Approve & publish

Reject this review

Feature on the homepage

CONTEXT

REVIEWER
Md. A.
arafatsarkar@mail.com

BOOKING
BK-2026-000012
Delivered —

SERVICE
Open Carrier Worldwide Transport

CARRIER
Lone Star Auto Logistics

DRIVER
Md. Arafat Driver

HELPFUL VOTES
0

MODERATED
Not yet

SUBMITTED FROM
127.0.0.1

Admin Reviews Good

Review by Md. A.

Review approved and published.

03F0B3Q66C7YTRSD8A3FK2W3 submitted 22 Aug 2026 12:45

Back to queue

THE REVIEW

Approved

★★★★★ 5.0 Verified purchase

Good

There is a Mark but it wiped off

Category ratings

COMMUNICATION 5.0 TIMELINESS 5.0

CONDITION 4.0 VALUE 4.0

Avg 4.50

PUBLIC REPLY

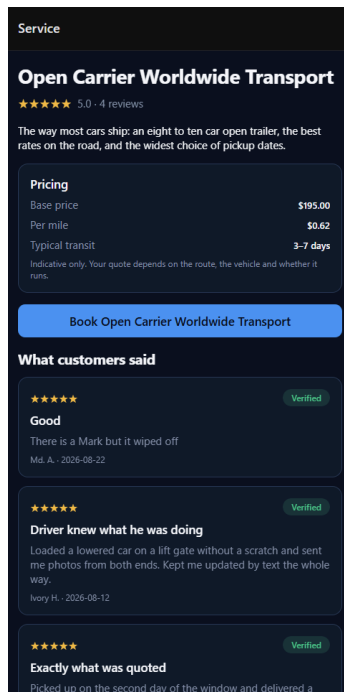
Write a reply

Published alongside the review on the public site.

Customers see this. Keep it short and specific.

Publish reply

5



Online Resources used:

a) Official documentation referred to during development:

22. Expo SDK 57 documentation — <https://docs.expo.dev/versions/v57.0.0/>
23. Expo Router (SDK 57) — <https://docs.expo.dev/versions/v57.0.0/sdk/router/>
24. Expo Router authentication and protected routes — <https://docs.expo.dev/router/advanced/authentication/>
25. Expo SecureStore — <https://docs.expo.dev/versions/v57.0.0/sdk/securestore/>
26. Laravel 12 documentation — <https://laravel.com/docs/12.x>
27. Laravel Sanctum (API token authentication) — <https://laravel.com/docs/12.x/sanctum>
28. Laravel validation and form requests — <https://laravel.com/docs/12.x/validation>
29. Laravel Eloquent relationships — <https://laravel.com/docs/12.x/eloquent-relationships>
30. React Native documentation — <https://reactnative.dev/docs/getting-started>
31. TanStack Query (server state for React Native) — <https://tanstack.com/query/latest>
32. Zod schema validation — <https://zod.dev>
33. MDN Web Docs (localStorage, Fetch, CORS) — <https://developer.mozilla.org/>

b) Open-source packages and repositories used:

34. spatie/laravel-permission — roles and permissions: <https://github.com/spatie/laravel-permission>
35. spatie/laravel-activitylog — admin audit trail: <https://github.com/spatie/laravel-activitylog>
36. laravel/sanctum — API tokens: <https://github.com/laravel/sanctum>
37. expo/expo — mobile framework and router: <https://github.com/expo/expo>
38. TanStack/query — data fetching and caching: <https://github.com/TanStack/query>
39. colinhacks/zod — client-side validation: <https://github.com/colinhacks/zod>
40. ulid/spec — sortable unique identifiers used for all public record ids: <https://github.com/ulid/spec>

Future Enhancements:

The following enhancements would extend the current system. Each one is listed with what already exists in the codebase, so the remaining work is clear.

41. Payment gateway integration (for example Stripe or SSLCommerz). The payment ledger, refund semantics and webhook table already exist; only the gateway calls are missing, so payments are currently recorded by staff by hand.
42. Email and SMS delivery. Contact-form replies are stored as an audit trail but are not sent, and password-reset mail is written to the log file in development.
43. Push notifications when a shipment changes status, so a customer does not have to open the app to check. The device token table and registration endpoint are already in place.
44. Live map tracking on the shipment screen. Driver position pings are already stored with latitude and longitude on the booking timeline.
45. Staff-issued quotes with versioning, and customer accept or decline in the app. The database supports versioned quotes; only the pricing form and the accept endpoint are outstanding.
46. Real distance lookup through a distance-matrix API, cached per lane, replacing the current fixed-distance estimate.
47. Claiming a guest quote request once the visitor registers and verifies the same email address.
48. Refunds and chargebacks exposed in the admin panel, rather than being ledger-only.
49. An automated test suite for the mobile client to match the backend coverage.